

(b) The distributive law of $+$ over $\cdot$ can be shown to hold true by means of a truth table.

x	y	z	$y \cdot z$	$x + y \cdot z$	$(x + y) \cdot z$	$(x + z) \cdot y$	$(x + y) \cdot (x + z)$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

5. From the complement table it is easily shown that:

(a) $x + x' = 1$, since $0 + 0' = 0 + 1 = 1$ and $1 + 1' = 1 + 0 = 1$

(b) $x \cdot x' = 0$, since $0 \cdot 0' = 0 \cdot 1 = 0$ and $1 \cdot 1' = 1 \cdot 0 = 0$ which verifies postulate

6. Postulate 6 is satisfied because the two-valued Boolean algebra has two distinct elements 1 and 0 with $1 \neq 0$.

We have just established a two-valued Boolean algebra having a set of two elements, 1 and 0, two binary operators with operation rules equivalent to the AND and OR operations, and a complement operator equivalent to the NOT operator.

2.3 Basic Theorems and Properties of Boolean Algebra

2.3.1 Duality

Duality principle: It states that every algebraic expression deducible from the postulates of Boolean algebra remains valid if the operators and identity elements are interchanged.

In a two-valued Boolean algebra, the identity elements and the elements of the set B are the same: 1 and 0. The duality principle has many applications. If the dual of an algebraic expression is desired, we simply interchange OR and AND operators and replace 1's by 0's and 0's by 1's.

Postulate 2:

(a) $x + 0 = x$

(b) $x \cdot 1 = x$

Postulate 5:

(a) $x + x' = 1$

(b) $x \cdot x' = 0$